

# Erdős Problem #373

EP-373

L1 - TRACTABLE

OPEN

Number Theory

## ORIGINAL PROBLEM STATEMENT

Show that the equation  $n! = a_1!a_2! \cdots a_k!$ , with  $n - 1 > a_1 \geq a_2 \geq \cdots \geq a_k \geq 2$ , has only finitely many solutions.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>